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Ensemble framework by using nature inspired algorithms for the early-stage forest fire rescue — A case study of dynamic optimization problems☆

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ABSTRACT

In this paper, we propose rescue ensemble to simulate the dynamic rescue process between forest fire spread and forest fire rescue, while simultaneously formulating this process as a dynamic optimization problem. However, there is still little research about simulating this kind of the dynamic rescue process, even when many new unmanned monitoring systems and large-scale firefighting aircraft emerge in the forest-fire-rescue field. Our rescue ensemble that consists of rescue simulator and rescue algorithm is characterized by supporting the offline simulation of the dynamic rescue process between forest fire spread (like offensive forces) and forest fire rescue (like defensive forces). Based on modifying the cellular automaton model of forest fire spread, rescue simulator is able to simulate forest fire spread and aircraft firefighting, simultaneously. Besides, the main goal of rescue algorithm is to realize the aircraft task allocation. Firefighting particle swarm optimization is proposed by us as our rescue algorithm, which is characterized by considering fire edge suppression, the burning-cell continuity, and wind direction. We construct our test problems based on real forest maps and aircraft firefighting capability. Comparing with four compared rescue algorithms, we test the different capabilities of firefighting particle swarm optimization, such as searching dynamic optimal solution, shortening the rescue time, controlling the spread speed of fire edge, and minimizing the burned cost. Experimental results demonstrate that the framework of rescue ensemble is feasible. Meanwhile, the results of firefighting particle swarm optimization are satisfactory in most cases.

1. Introduction

Destructive disasters lead to heavy casualties and economic losses, such as earthquakes, tsunamis, hurricanes, floods, mud-rock flows, and fires. Emergency responses are critical to the disaster-rescue success (Sheu, 2007). However, rescue resource allocation usually belongs to NP-hard problems (Simpson and Hancock, 2009; Wex et al., 2014). In addition, disaster rescues need to satisfy the special challenges for each kind of disasters, because of the different features of disasters. The special challenge of forest fire rescue is the dynamic features of forest fire spread and the quick response requirements.

Critical problems are found out in the early forest-fire-rescue studies (Martell, 1982) and the important conclusions are put forward (Parlar and Vickson, 1982; Parlar, 1983; Hobeika et al., 1988; Brown and Vassiliou, 1993; Alonso-Betanzos et al., 2003; Chi et al., 2003; Rachaniotis and Pappis, 2006; Daugherty and Fried, 2007). Comparing with the early technical platforms, unmanned platforms have been widely used in many industries, such as pilotless bus (Sohu, 2019A), driverless car (Sohu, 2019B), unmanned rescue (Wiki, 2019), and so on. In the latest decade, many new equipments have emerged, such as large firefighting aircraft (People, 2019; Weixin, 2019), firefighting helicopter with bucket (Wangyi, 2019; Sohu, 2019C), firefighting unmanned aerial vehicle (UAV) (Baidu, 2019), new unmanned monitoring systems (Eastday, 2019), and so on. Meanwhile, there are new aircraft researches of forest fire rescue (Snyder et al., 2012; Marchi et al., 2014; Karma et al., 2015) and new monitoring systems are able to provide forest fire situations in real time (Eastday, 2019; Sohu, 2019D; Ollero et al., 2006; Yuan et al., 2015). For example, AG600 is able to load 12 tons of water as large-scale firefighting aircraft (People, 2019; Weixin, 2019) and the bucket of new firefighting helicopter is able to load 1.6 tons of water (Wangyi, 2019). These new aircraft are changing our emergency response procedure (Sohu, 2019D) and method (Ollero et al., 2006; Yuan et al., 2015) for forest fire rescue. Generally, these provide an opportunity to realize aircraft task allocation in the early-stage forest fire rescue, especially for dealing with the dynamic feature

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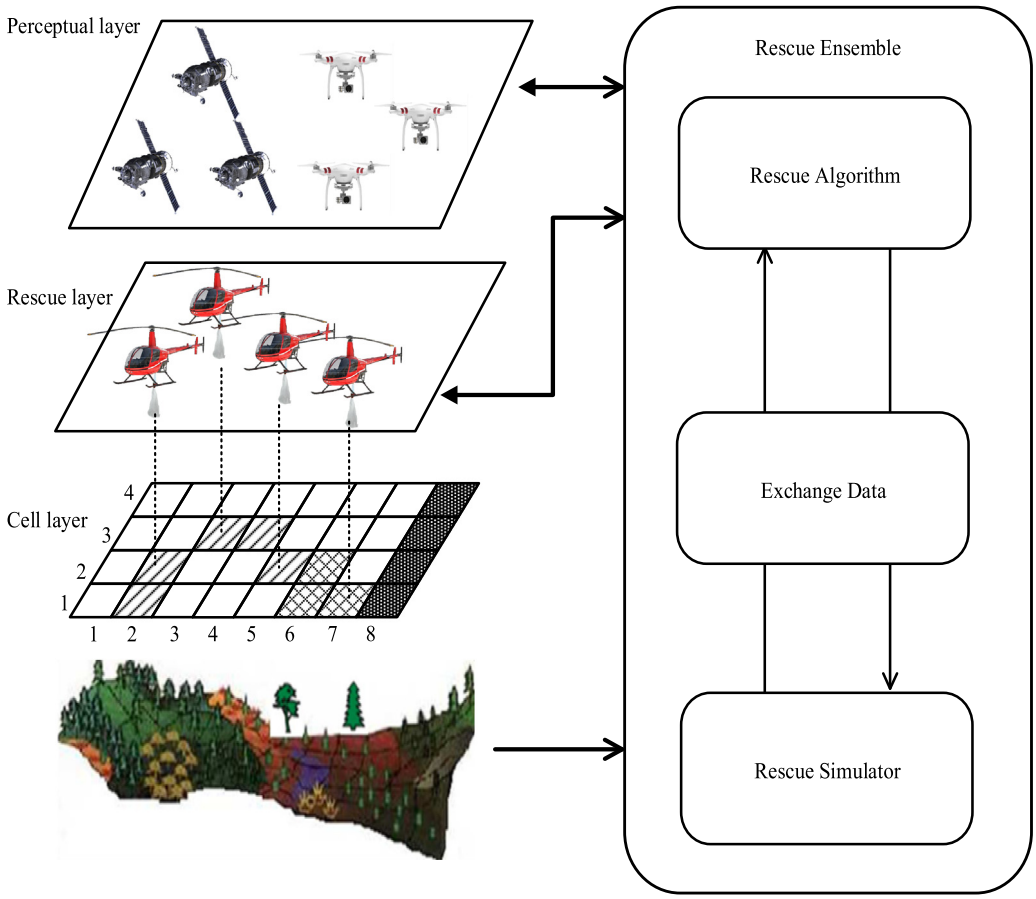


Fig. 1. Our problem focuses on the aircraft task allocation in the early-stage forest fire rescue. Note that, rescue ensemble includes rescue simulator and rescue algorithm.

of forest fire spread. How to solve aircraft task allocation for forest fire rescue is an important problem.

We focus on new aircraft task allocation in the early-stage forest fire rescue, as shown in Fig. 1. There are three layers in Fig. 1. In the perceptual layer, remote sensing satellites and UAVs are able to provide the specific data of forest fire. In the rescue layer, firefighting helicopters or UAVs are used to put out forest fire. In the cell layer, we obtain the map and vegetation information of forest fire and then construct 2-D grid map with cells. Besides, rescue ensemble realizes exchanging data between rescue simulator and rescue algorithm in Fig. 1. The early-stage forest fire rescue is important to control forest fire spread and put out forest fire as soon as possible. We only consider the early-stage forest fire rescue in this paper and some assumptions are as follows. (i) There are only aircraft and no other firefighting vehicles (or devices) that participate in the early-stage forest fire rescue. The reason is that aircraft have rapid speed and first arrive. Besides, the aircraft firefighting equipment is the bucket to load water and spray water. (ii) We ignore the flight-altitude change of rescue aircraft and compute the flight distance of aircraft in 2-D grid map. The reason is that the total region of forest fire spread in the early stage is usually small (like 1.5 km × 1.5 km in our test maps or even smaller) and then the flight-altitude change of rescue aircraft is usually small. Meanwhile, we only consider forest fire rescue in 1.5 km × 1.5 km fire-site region and do not consider other regions (like exceeding 1.5 km × 1.5 km fire-site region). (iii) Surface fire is a common kind of forest fire. We assume that there is only surface fire in the early-stage forest fire and do not consider other kinds of forest fire, such as spotting fire.

There are some difficulties to solve this problem as follows. Difficulty I. Aircraft task allocation for forest fire rescue belongs to dynamic optimization problems (DOPs) because forest fire spread is dynamic with uncertainty. There are many kinds of forest fire spread models in Section 2.1. However, there is little research about rescue simulators in the forest fire field, as we know. Difficulty II. There is little research about the framework of integrating forest fire spread with forest fire rescue in Section 2.2 and this framework is able to simulate the dynamic rescue process between forest fire spread (like offensive forces) and forest fire rescue (like defensive forces), as we know.

To overcome these difficulties, we propose the framework of our rescue ensemble. Our contributions are as follows. (i) We formulate aircraft task allocation in the early-stage forest fire rescue as a DOP and this DOP is characterized by considering the dynamic features of forest fire spread and rescue. Based on 2-D grid map, our model includes aircraft tasks, aircraft mobility, aircraft firefighting load, and aircraft firefighting-load supplement. (ii) To overcome Difficulty I, we introduce rescue simulator based on modifying the cellular automaton (CA) model of forest fire spread. To overcome Difficulty II, rescue ensemble provides the framework to coordinate the work between rescue simulator and rescue algorithm. Besides, nature inspired algorithms are widely used because of their good performances, such as particle swarm optimization (PSO) (Kennedy and Eberhart, 1995), modified PSO (Shi and Eberhart, 1998), augmented Lagrangian PSO (Shahri et al., 2019), imperialist competitive algorithm (Atashpaz-Gargari and Lucas, 2007), enhanced shuffled frog-leaping algorithm (Liu et al., 2015), gases Brownian motion optimization (Abdechiri et al., 2013), bacterial foraging optimization (Pang et al., 2019), fractional-order firefly algorithm (Mousavi and Alfi, 2018), and so on. Therefore, we propose firefighting PSO (FPSO) as rescue algorithm, while simultaneously introducing the coding method to realize the control of forest fire edge. (iii) To test rescue simulator, rescue ensemble, and FPSO, our benchmarks used real forest maps. Comparing with four compared rescue algorithms, the FPSO results about forest-fire-rescue time (*FT*), rescued edge ratio (*RER*), and burned value ratio (*BVR*) were satisfactory.

Table 1
Summary of forest fire spread models. Besides, SF, MF, and FA mean surface-fuel-layer, multiple-fuel-layer, and fuel-layer & atmosphere radiations, respectively.

Model	Radiation	Main method description	Factors					
			Fuel type	Wind	Humidity	Temperature	Slope	Fuel moisture
PYROCART model (Perry et al., 1999)	SF	Rothermel fire spread model	√	√			√	√
Hexagonal CA model (Encinas et al., 2007)	SF	Hexagonal cellular automata model	√	√			√	
Elliptical model (Glasa and Halada, 2008)	SF	Elliptical model of forest fire spread based on classical envelope theory		√			√	
CA model (Alexandridis et al., 2008)	SF	Cellular automata and statistical model	√	√			√	
Multi-dimension Geo-CA model (Sun et al., 2013)	SF	Wang Zhengfei fire spread velocity model	√	√	√	√	√	
Crown model (Cruz and Alexander, 2017)	MF	Monte Carlo simulation framework for nonlinear fire behavior	√	√	√	√		√
Crown experiment (Silva et al., 2017)	MF	Full-scale crown fire experiment	√	√		√	√	√
CA & ELM model (Zheng et al., 2017)	SF	Cellular automaton with extreme learning machine based on local fire historic training data		√				
FIREFLY model (Zhang et al., 2017)	SF	Data-driven wildland fire spread model based on data assimilation algorithm	√	√			√	√
Two-way coupling model (Lopes et al., 2017)	FA	Two-way coupling method based on Rothermel model	√	√			√	

The rest of the paper is organized as follows. In Section 2, we review related work. In Section 3, we formulate our problem. In Sections 4, 5, and 6, we introduce our rescue ensemble, rescue simulator, and FPSO rescue algorithm, respectively. In Section 7, we give experimental results and discussions. In Section 8, we present our findings and future work.

2. Related work

2.1. Forest fire spread models

There is little research about forest fire rescue models, as we know. In the forest fire field, most of the researches are forest fire spread models (Sullivan, 2009; Brun et al., 2014; Thomas et al., 2017). Therefore, we only summarize the features of forest fire spread models in Table 1.

2.2. Forest fire rescue and management

The kernel objective of forest fire rescue and management is to realize fire suppression as soon as possible.

Firstly, from a deteriorating job perspective, Parlar (1983) focuses on optimal forest fire control with limited reinforcements and proposes a generalized forest firefighting model that is solved by the maximum principle. Rachaniotis and Pappis (2006) consider scheduling a single firefighting resource and tackle this problem by using the concept of deteriorating jobs. Kali (2016) concerns the optimal deployment of a single forest suppression processor and uses stochastic approach to capture the uncertainty of forest fire.

Secondly, there are some important researches about decision support systems as follows. Brown and Vassiliou (1993) propose a real-time decision support system that involves optimization method, simulation, and decision-maker judgment. Alonso-Betanzos et al. (2003) present a comprehensive management system that is able to predict forest fire risk, back up the forest fire monitoring & extinction phase, and assist in planning the recuperation of the burned areas. Chi et al. (2003) develop an efficient decision support system whose framework is introduced in detail.

Lastly, there are two kinds of objective-oriented rescue researches as follows. (i) Single objective fire management. Daugherty and Fried (2007) present a joint-optimization approach that simultaneously selects acres to be treated by fuel treatment prescription and assigns bioenergy production facility locations and capacities. Rytwinski and Crowe (2010) use a fuel break approach to restrict the spread of wildfire and design a stochastic simulation model of fire spread to compute the fire risk of each region. Ren and Tian (2016) consider minimizing the total travel distance of firefighting teams and formulate the emergency scheduling model. (ii) Multi-objective scheduling rescue algorithms. Ren et al. (2015) formulate multi-objective scheduling model to minimize the total rescue time and the amount of fire rescue resource, simultaneously. Tian et al. (2016) propose a novel multi-objective scheduling model and introduce a fire-spread speed model to describe practical forestry fire. Wu et al. (2018) use precise dynamic programming algorithms in small-scale fires and use multi-objective integer programming to achieve multi-depot fire engine scheduling.

In summary, the strategic decision objective is to extinguish forest fire as soon as possible, especially in the early-stage forest fire. Extinguishing forest fire immediately means the minimal cost loss (like the burnt-forest-cost loss) and the maximal social efficiency (like the safety of firemen). On the other hand, the tactical decision objective is to control the spread edge of forest fire. Controlling the spread edge of forest fire effectively means that extinguishing forest fire is possible. Otherwise, it is impossible.

In essence, forest fire spread and rescue are dynamic time-varying and this problem belongs to DOPs. Moreover, considering forest fire spread and rescue as a DOP is suitable for real forest-fire-rescue requirements. However, there is little research about forest fire spread and rescue as a DOP, as we know. Thus, we do some exploratory researches to address forest fire spread and rescue as a dynamic task allocation problem in this paper.

3. Problem statement and formulation

3.1. Statement

We formulate the dynamic process of forest fire rescue as aircraft task allocation problem to minimize the flight time and the newly-increased burnt forest cost, simultaneously. Besides, our model is also able to provide *FT*, *RER*, and *BVR*.

Table 2
Definition in rescue ensemble, rescue simulator, and rescue algorithm.

Type	Subtype	Symbols	Descriptions
Rescue ensemble	Time	t	Simulation time
		T_{ra}	The time period of running rescue algorithm once
		T_{rs}	The time period of running rescue simulator once
		T_{res}	Time resolution, $T_{res} = min(T_{ra}, T_{rs})$, and $min()$ returns the minimum value
	Rescue aircraft	NT	The number of all aircraft, and $NT = N0+N1+N2$.
		AID	Aircraft's ID, and $AID = u(1 \le u \le NT)$
		FEL	The maximum firefighting load of each aircraft
		FEC	The maximum number of firefighting cells of each aircraft, which is decided by FEL and the cell area.
		L^t	The firefighting load vector of all aircraft $L^t = [l_1^t, l_2^t, \ldots, l_u^t, \ldots, l_{NT}^t]$ at t and l_u^t is the firefighting load of the uth aircraft at t.
		V	The flight speed of each aircraft
Rescue simulator	Map	$cell$	A cell in 2-D grid map
		L	The length and width of each cell
		i	The horizontal coordinate i, $1 \le i \le R$ (i.e., the cell number of each row)
		j	The vertical coordinate j, $1 \le j \le C$ (i.e., the cell number of each column)
		CID	Cell's ID, which is equal to $i + (j-1) \times C$.
	Cell states, rules, and state matrix	$S1$–$S5$	Cell states. $S1$: incombustible region. The cells contain no forest, and these cells in $S1$ state cannot be burned. $S2$: forest with no fire. The cells contain forest, which has not been ignited yet. $S3$: burning forest. The cells contain forest, which is burning. $S4$: burned-down forest. The cells contained forest, which has been burned down. $S5$: rescued forest. The cells were burning. After rescue, the cells' fire has been put out.
		$R1$–$R7$	Rules in rescue simulator
		$\boldsymbol{SM}$	State matrix records the states of all cells in 2-D grid map.
	Burnt cost	$Value\ (i, j)$	The value of each cell $Value\ (i, j) \in \{1, 2, 3\}$. These values correspond to sparse, normal and dense vegetation densities, respectively. Meanwhile, the cost value level of each cell is directly proportional to the density of forest vegetation.
		$NewBurntSet^T$	The cell set of the newly-increased burnt cells in one T_{rs}. We suppose that t_0 is the beginning time of one T_{rs} of rescue simulator. Let $BeginTimeS3SET$ be the cell set with burning cells (i.e., cell state $= S3$) at t_0. Let $EndTimeS4SET$ be the cell set with burnt cells (i.e., cell state $= S4$) at $t_0 + T_{rs}$. Then, $NewBurntSet^T = BeginTimeS3SET \cap EndTimeS4SET$.
	Aircraft group I	$N0$	The number of all aircraft in rescue simulator
Rescue algorithm	Aircraft group II	$N1$	The number of aircraft group II with $l_u^t = FEL$ in rescue algorithm
		XP^t	The latitude & longitude positions of aircraft group II at t, and $XP^t = [xp_1^t, xp_2^t, \ldots, xp_k^t, \ldots, xp_{N1}^t]$. xp_k^t is the position of the kth aircraft with $l_u^t = FEL$.
		X^t	The firefighting destination cells of aircraft group II at t, and $X^t = [x_1^t, x_2^t, \ldots, x_k^t, \ldots, x_{N1}^t]$. x_k^t is the firefighting destination cell of the kth aircraft.
	Aircraft group III	$N2$	The number of aircraft group III with $l_u^t = 0$ in rescue algorithm
		YP^t	The latitude & longitude positions of aircraft group III at t, and $YP^t = [yp_1^t, yp_2^t, \ldots, yp_q^t, \ldots, yp_{N2}^t]$. yp_q^t is the position of the qth aircraft with l_u^t =0.
		Y^t	The nearest firefighting bases of aircraft group III, and $Y^t = [y_1^t, y_2^t, \ldots, y_q^t, \ldots, y_{N2}^t]$. y_q^t is the nearest firefighting base of the qth aircraft.

3.2. Definition

See Table 2.

3.3. Formulation

We formulate our model (1) to minimize the longest flight time of aircraft group II and the newly-increased burnt forest cost in one period, simultaneously. a ∈ [0, 1], $TMAX$ is the flight time of the diagonal line in a grid map, $CMAX$ is approximately equal to the forest cost in the outer edge cells of a grid map, $LCM()$ returns the least common multiple, $LCM(CMAX, TMAX)$ is used to unify the flight time and the newly-increased burnt forest cost, and $max()$ returns the maximal value. $CMAX$ is used as a normalized parameter, because we assume the worst situation of the newly-increased burnt forest cost in one T_{rs} is that the outer edge cells of a grid map are burning (cell state $= S3$). Meanwhile, Eq. (1) is also the fitness function of rescue algorithms, and α adjusts the proportion between the flight time and the burnt forest cost. $\alpha = 0.5$ in this paper (see Table 3).

$$F\left(X^t\right) = \alpha \frac{CMAX}{LCM\left(CMAX, TMAX\right)} \cdot MaxTime\left(X^t, XP^t\right) + (1-\alpha)\frac{TMAX}{LCM\left(CMAX, TMAX\right)} \cdot BurntCost\left(X^t, XP^t\right) \quad (1)$$

$$TMAX = L\sqrt{R^2 + C^2}/V \quad (2)$$

$$CMAX = \frac{\sum_{j=1}^{C}\sum_{i=1}^{R} Value(i,j)}{R \times C} \times (2R + 2C - 4) \quad (3)$$

$$MaxTime\left(X^t, XP^t\right) = max(\{\left\|\frac{x_1^t - xp_1^t}{v}\right\|, \left\|\frac{x_2^t - xp_2^t}{v}\right\|, \ldots,$$

Table 3
Rules in our rescue simulator. t_0 is the beginning time of one T_{rs}. Then, the next time of performing rescue simulator is $t_0 + T_{rs}$. The state (i, j, t_0) of cell (i, j) at t_0 may be $S1$–$S5$. Besides, $S1$ means incombustible region, $S2$ means forest with no fire, $S3$ means burning forest, $S4$ means burned-down forest, and $S5$ means rescued forest. If aircraft $(i, j, t_0) = 0$, this means that there is no aircraft in the cell (i, j) at t_0. Otherwise, aircraft $(i, j, t_0) = 1$.

No.	Content	Meaning
R1	If state $(i, j, t_0) = S1$, then state $(i, j, t_0 + T_{rs}) = S1$.	The state of no vegetation cells remains the same.
	R2A: If state $(i, j, t_0) = S3$ and aircraft (i, j, t_0)=0, then state $(i, j, t_0 + T_{rs}) = S4$	If aircraft is not arranged to extinguish the burning cell at t_0, the cell will be burned down at $t_0 + T_{rs}$.
R2	OR	OR
	R2B: If state $(i, j, t_0) = S3$ and aircraft (i, j, t_0)=1, then state $(i, j, t_0 + T_{rs}) = S5$.	If aircraft is arranged to extinguish burning cell at t_0, the fire will be put out at $t_0 + T_{rs}$.
R3	If state $(i, j, t_0) = S4$, then state $(i, j, t_0 + T_{rs}) = S4$.	The state of the burned cells remains unchanged. That is to say, the burned cells cannot be rekindled.
R4	When state $(i, j, t_0) = S3$ and aircraft $(i, j, t_0) = 0$, if state $(i \pm 1, j \pm 1, t_0) = S2$, then state $(i \pm 1, j \pm 1, t_0 + T_{rs}) = S3$ with the p_{burn} probability in Eq. (6).	If a cell is on fire at t_0 and there is no aircraft, the fire can spread to neighboring cells with the p_{burn} probability at $t_0 + T_{rs}$. Note that, in respect of each state $(i \pm 1, j \pm 1, t_0 + T_{rs})$, we generate a random number $\lambda \in [0, 1]$. If $\lambda < p_{burn}$, then this state = $S3$. Otherwise, this state is still *S2*.
R5	When state $(i, j, t_0) = S3$ and aircraft $(i, j, t_0) = 1$, if state $(i \pm 1, j \pm 1, t_0) = S2$, then state $(i \pm 1, j \pm 1, t_0 + T_{rs}) = S2$.	If a cell is on fire at t_0 and there is aircraft, then the fire cannot spread to neighboring cells
R6	If state $(i, j, t_0) = S5$, then state $(i, j, t_0 + T_{rs}) = S5$.	The state of the extinguished cells remains unchanged. That is to say, the cells that have been extinguished cannot be rekindled.
R7	If state $(i, j, t_0) = S2$ and state $(i \pm 1, j \pm 1, t_0) \neq S3$, then state $(i, j, t_0 + T_{rs}) = S2$	If the neighbors of a cell with state = $S2$ are not on fire, then this cell's state is still *S2*.

$$\left\| \frac{x_k^t - xp_k^t}{v} \right\|, \ldots, \left\| \frac{x_{N1}^t - xp_{N1}^t}{v} \right\| \}) \tag{4}$$

$$BurntCost\left(X^t, XP^t\right) = \sum_{cell \in NewBurntSet^T} Value(i, j) \tag{5}$$

The features of our model are as follows. (i) Our aircraft task allocation problem belongs to DOPs. The reason is that forest fire spread is dynamic time-varying and explanations are as follows. In real forest fire, the shape of forest fire is dynamic time-varying. Similarly, forest fire spread corresponds to the shape of forest fire and has uncertainty in our rescue simulator. As shown in the rule *R4*, rescue simulator simulates this kind of uncertainty (i.e., that p_{burn} is greater than or less than a random number λ). This leads to the uncertainty of cell states and results in the time-varying shape of forest fire. Therefore, the shape of forest fire in rescue simulator is also dynamic time-varying. Thus, $MaxTime(X^t, XP^t)$ and $BurntCost(X^t, XP^t)$ are time-varying. (ii) X^t is not only the firefighting destination cells of aircraft group II, but also the individual coding in rescue algorithms. Meanwhile, *N1* is the individual coding length that also changes with time increasing.

The limits of our model are as follows. (i) Aircraft is homogeneous in our model. This means that each aircraft has the same *FEL*, *FEC*, and *V*. If aircraft is heterogeneous, these scalars should be replaced with the new vector definitions. (ii) In each T_{ra}, the task allocation solution of rescue algorithm should be transmitted to each aircraft at once. Thus, our model requires that the communication links between base stations and aircraft must always be maintained.

4. Rescue ensemble

Rescue ensemble consists of rescue simulator and rescue algorithm, which is able to support the initialization and running processes of rescue simulator and rescue algorithm.

4.1. Framework

The framework of rescue ensemble is given in Algorithm 1 and Fig. 2. Algorithm 1 mainly describes updating aircraft groups' members and exchanging data.

The termination conditions of rescue ensemble in Algorithm 1 are different from the stop condition of rescue algorithm in Algorithm 5. (i) Termination condition I is that there are no burning cells in a grid map. This corresponds to two situations. One is that aircraft put out all fires. The other is that all cells have been burnt down. (ii) Termination condition II is that *RunNum* is greater than or equal to 200. As shown in *Step 4*, the simulation time of running rescue algorithm 200 times is equal to the simulation time of updating ***SM*** 100 times in this paper. Thus, Termination condition II means that our limited aircraft (i.e., *NT* = 10 in this paper) is not able to realize fire suppression, because the 100 times of forest fire spread (i.e., the 100 times of updating ***SM***) result in so many burning cells in forest fire edge.

Our simulation does not synchronize with the real forest fire. That is to say, the simulation time t is not synchronized with the actual clock time in our lives. Thus, our simulation belongs to offline simulation and our ensemble framework is not able to support online simulation. To support online simulation, some modifications are as follows. Rescue ensemble should replace rescue simulator with the real system of forest fire situation that may be obtained by remote sensing satellites or UAVs. Besides, T_{ra} and the period of updating forest fire situation all change over time. Therefore, the time points of exchanging data between rescue algorithm and the real system of forest fire situation should support this requirement. Generally, we believe that the online system of forest fire rescue is an emerging and interesting research in the future.

4.2. Update aircraft groups' members

The process of updating aircraft groups' members is explained in Fig. 3 and Algorithm 2. Generally, each aircraft changes from aircraft group I to aircraft group III, from aircraft group III to aircraft group II, and from aircraft group II to aircraft group I, repeatedly. Besides, this process also explains the rescue simulation principle of rescue ensemble by using aircraft as a clue.

4.3. Exchange data

Fig. 4 is the example of exchanging data between rescue simulator and rescue algorithm to demonstrate our paper's settings. Explanations are as follows. (i) The time points of exchanging data are the least common multiple of T_{rs} and T_{ra}. Therefore, rescue ensemble is able to support various values of T_{rs} and T_{ra}. (ii) The kinds of the exchanged data are as follows. In the beginning of each T_{rs}, rescue ensemble updates aircraft groups' members and rescue simulator updates ***SM***. In the beginning of each T_{ra}, rescue ensemble updates aircraft groups' members and rescue algorithm provides its aircraft task allocation solution.

Algorithm 1: Rescue ensemble

Step 1: According to the initial ignition point, initialize ***SM*** by running forest fire spread simulator (Alexandridis et al., 2008) 10 times. Meanwhile, to reduce the difference of initializing ***SM*** in each test, p_{burn} is equal to $0.8+0.2\times rand$ ($rand \in [0, 1]$) in this fire spread simulator (see Eq. (6)). Besides, initialize the states of all aircraft and their groups.

Step 2: Define the number of running rescue algorithm as *RunNum* and initialize *RunNum* equal to 0. Note that, $T_{res}=min(T_{ra}, T_{rs})=T_{ra}$, $T_{rs}=2T_{ra}=20$s, and $T_{ra}=10$s in this paper. Then, the simulation time of running rescue algorithm once is T_{ra} (i.e., 10s in this paper).

Step 3: If *N1* is not equal to 0 (Note that, *N1* is always not equal to 0 in this step of our paper), execute rescue algorithm in Algorithm 5, output its aircraft task allocation solution, and update *RunNum* by plus one.

Step 4: By using the aircraft task allocation solution of rescue algorithm as input, execute rescue simulator in Algorithm 3. Note that, in Algorithm 3, when the simulation time is not an integer multiple of T_{rs}, only update aircraft groups' members by using Algorithm 2. Otherwise, simultaneously update aircraft groups' members and ***SM***. This process of exchanging data is given in Fig. 4. That is to say, the simulation time of updating ***SM*** once is T_{rs} (i.e., 20s in this paper). Generally speaking, according to $T_{rs}=2T_{ra}$, the simulation time of updating ***SM*** once is equal to the simulation time of running rescue algorithm twice in this paper.

Step 5: If the termination conditions of rescue ensemble are not satisfied, update aircraft groups' members by using Algorithm 2 and go to *Step 3*. Otherwise, stop.

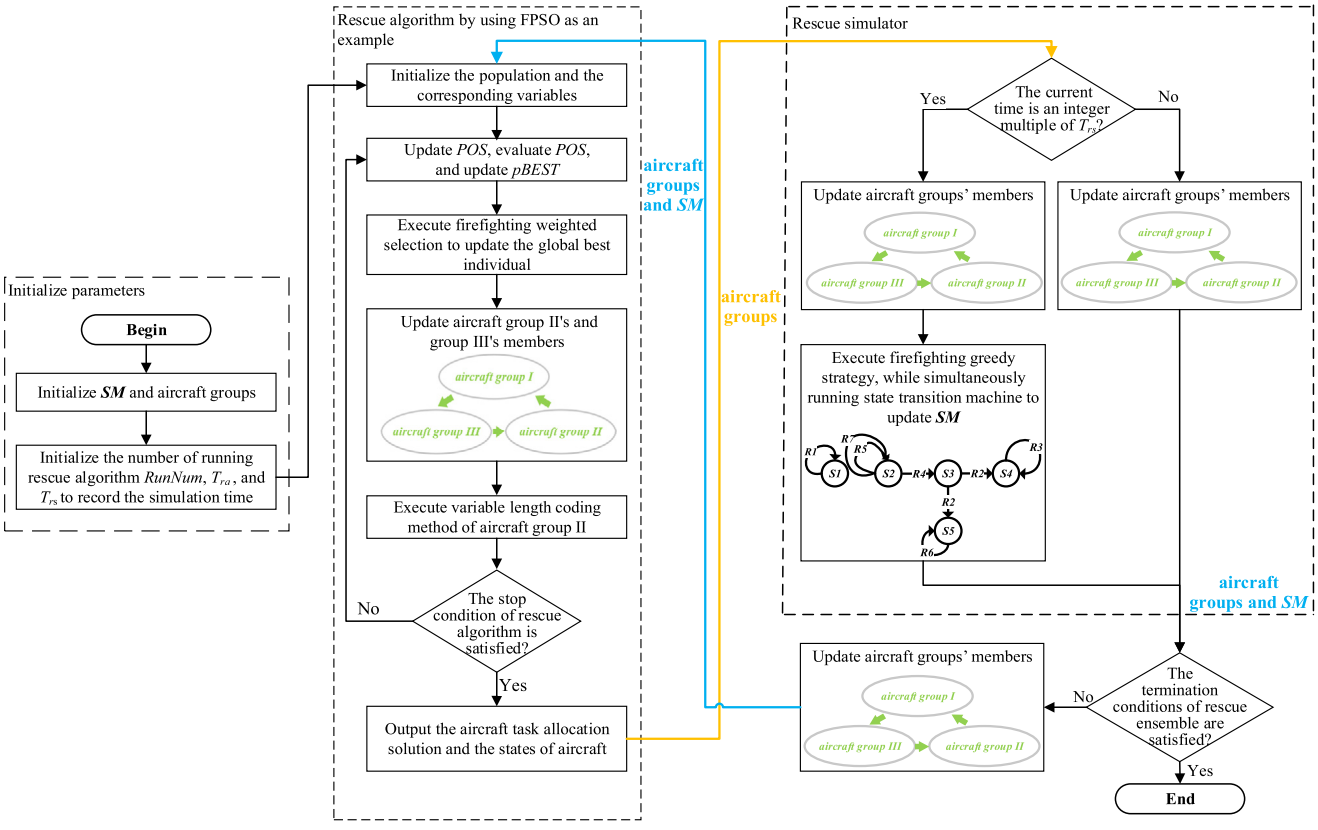


Fig. 2. An illustration of our rescue ensemble that focuses on the offline simulation of the dynamic rescue process between forest fire spread (like offensive forces) and forest fire rescue (like defensive forces). Besides, this figure includes important input, output, and main steps by using FPSO as an example.

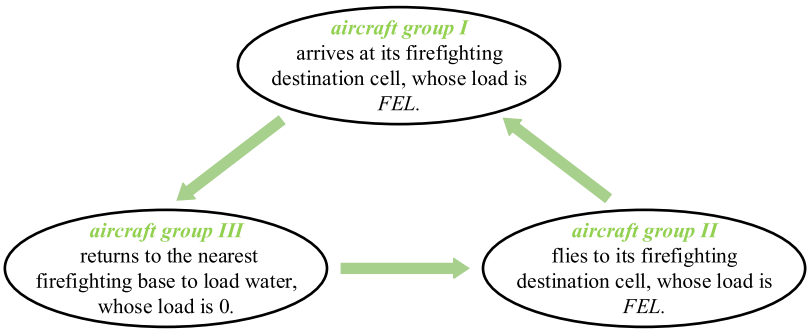


Fig. 3. The process of updating aircraft groups' members to demonstrate the rescue process of each aircraft.

Algorithm 2: update aircraft groups' members

Input: the load & location of each aircraft (i.e., L^t, XP^t, YP^t), X^t, Y^t, ***SM***, $N0$, $N1$, and $N2$

Output: the updated members of aircraft groups I-III, the updated values of $N0$, $N1$, and $N2$

Step 1: For each aircraft in aircraft group I, if the aircraft's load changes from *FEL* to 0 (i.e., this means that the aircraft begins to spray water and then its load is 0), copy this aircraft into aircraft group III.

Step 2: For each aircraft in aircraft group III, if the aircraft's load changes from 0 to *FEL* (i.e., this means that the aircraft return to the nearest firefighting base to load water and then its load is *FEL*), copy this aircraft into aircraft group II.

Step 3: For each aircraft in aircraft group II, if the aircraft arrives at its firefighting destination cell, copy it into aircraft group I.

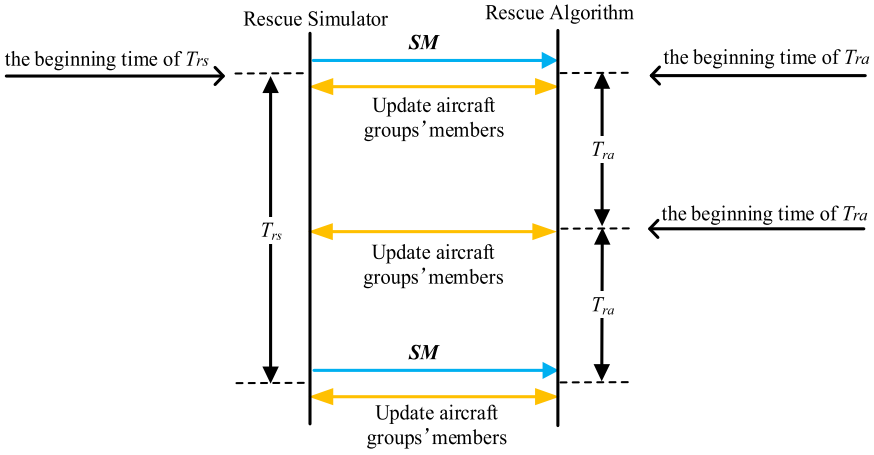


Fig. 4. An example of exchanging data between rescue simulator and rescue algorithm to demonstrate our paper's settings (i.e., $T_{ra} = 10$ s, $T_{rs} = 20$ s, and $T_{rs} = 2T_{ra}$).

5. Rescue simulator

Rescue simulator is able to simulate forest fire spread and rescue, simultaneously. This is different from forest fire spread models in Section 2.1.

5.1. Overview

Rescue simulator is the same with forest fire spread simulator in Alexandridis et al. (2008), except for minor modifications as follows. (i) We do not consider spotting effect (corresponds to spotting fire) to simplify forest-fire-spread simulation, as we assumed in Section 1. (ii) We add state *S5* and *S5* means that the cells' fire has been put out. Meanwhile, we only modify *R2* and *R6* rules, because of using aircraft. (iii) Aircraft group I uses the firefighting greedy strategy.

5.2. Cell automaton

In rescue simulator, vegetation type, vegetation density, wind speed, wind direction, and altitude not only affect the shape of forest fire, but also affect the spread speed of forest fire.

5.2.1. Rule

In Alexandridis et al. (2008), the probability p_{burn} of *R4* is given by

$$p_{burn} = p_h \left(1+p_{veg}\right) \left(1 + p_{den}\right) p_w p_s \quad (6)$$

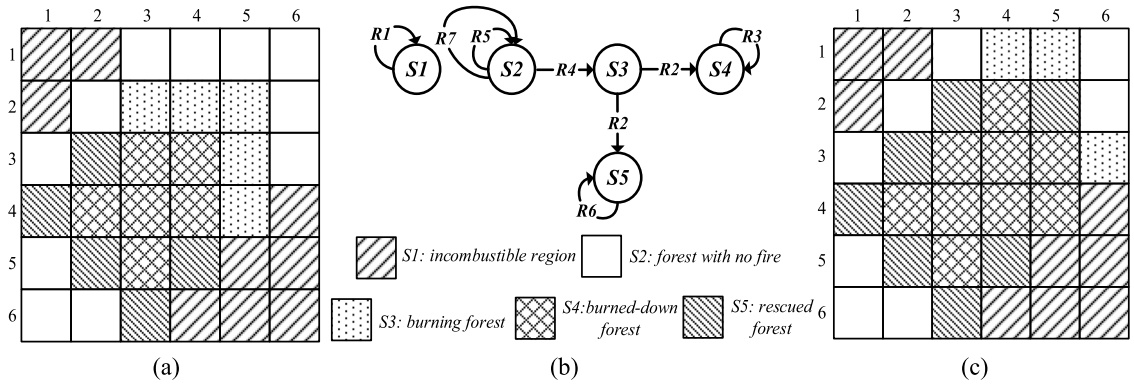


Fig. 5. An example to demonstrate the visual process of a state transition. (a) is the map of fire spread (corresponds to ***SM***) at the beginning time t_0, (b) is a state machine, and (c) is the map of fire spread at the next time $t_0+\mathrm{T}_{rs}$.

where p_h is the constant probability that a cell adjacent to a burning cell containing a given type of vegetation and density will catch fire at the next time step under no wind and flat terrain. In this paper, $p_h = 0.58$. p_{veg}, p_{den}, p_w, and p_s are the fire propagation probabilities that depend on the type of vegetation, the density of vegetation, the wind speed, and the slope, respectively. Details are as follows. (i) Effect of vegetation type and density. The influence of vegetation type and density is expressed by probability p_{veg} and p_{den}, respectively. The vegetation types are divided into agricultural areas, shrubs forest, and pine forest. Meanwhile, the corresponding p_{veg} values are −0.3, 0, and 0.4 respectively. In this paper, $p_{veg} = 0$. The density of vegetation can be divided into sparse, normal, and dense vegetation densities. Meanwhile, the corresponding p_{den} values are −0.4, 0, and 0.3, respectively. (ii) Effect of wind speed and direction. The wind's influence on the fire spread includes the empirical wind–effect relationship between wind speed and wind direction, which is given by

$$p_w = exp\left(c_1 WV\right) f_t \tag{7}$$

$$f_t = exp(WV c_2(\cos\theta - 1)) \tag{8}$$

where $c_1 = 0.045$ and $c_2 = 0.131$. θ is the angle between the fire propagation direction and the wind direction, which is from 0° to 360°. WV is wind speed. Note that, in our benchmarks, there are various wind-speed levels, which correspond to different wind speeds WV. (iii) Effect of altitude. The altitude's effect on the fire propagation speed p_s is equal to $exp(a\theta_s)$, a is a setting parameter ($a = 0.078$ in this paper), and θ_s is the slope angle. When a cell is an adjacent cell, θ_s is equal to $\tan^{-1}((E_1 - E_2)/l)$. E_1 and E_2 are the altitudes of the two cells, and l is the length of the square cell. When a cell is a diagonal cell, the slope angle is equal to $\tan^{-1}((E_1 - E_2)/1.414l)$.

5.2.2. State transition

An example to demonstrate the visual process of a state transition is provided in Fig. 5. In our map, each cell provides eight possible directions of fire propagation. Explanations are as follows. (i) *R1*: See (1, 1) cell. (ii) *R2*: If aircraft are not arranged at cells in state *S3*, then these cells' states will be converted to *S4* (see (2, 4) cell). Otherwise, if aircraft are arranged at some cells in state *S3* at *t*, then the state of these cells will be converted to state *S5* (see (2, 3) cell). (iii) *R3*: See (5, 3) cell. (iv) *R4* and *R5*: In respect of the cells in state *S2* and surrounding cells that have ignition ability, we compute the probability p_{burn} and generate a random number $\lambda \in [0, 1]$. If $\lambda < p_{burn}$, then converts these cells' states from state *S2* to state *S3* (see (1, 4) cell). Otherwise, these cells' states remain unchanged (see (1, 3) cell). (v) *R6*: See (4, 1) cell. (vi) *R7*: See (5, 1) cell.

5.3. Procedure

Rescue simulator is given in Algorithm 3.

Algorithm 4 introduces the firefighting greedy strategy of aircraft group I. Besides, all aircraft use the buckets to load water in this paper. Thus, we assume that each aircraft accomplishes spraying water only in one T_{rs}. Note that, there is a situation that more than one aircraft simultaneously reach the same firefighting destination cell. In this kind of situation, the firefighting strategy of these aircraft is the same with Algorithm 4, except that these aircraft are not allowed to repeat firefighting for the same burning cell.

We explain the validity and reasonability of the aircraft firefighting strategy as follows. (i) On validity. The size of cells (L) is able to be artificially adjusted. Reducing L gradually is effective to simulate the firefighting reality of aircraft. The smaller L, the higher validity. (ii) On reasonability. The key factor to put out the fire in one cell is the sprayed water load per unit area (like 60 kg/m^2). If there is enough water in one burning cell, it is certain that the aircraft is able to put out the fire in this cell. Note that, *FEC* is decided by *FEL* and L in this paper. We explain real data (Wangyi, 2019; Sohu, 2019C; People, 2019; Weixin, 2019) of *FEC* and *FEL* in Section 7.1.1.

6. FPSO rescue algorithm

In this section, we use FPSO as an example to demonstrate the function and realization scheme of rescue algorithms.

6.1. Motivation

The motivation of FPSO is to realize the task allocation of aircraft group II for forest fire rescue in an effective and efficient way.

6.2. Framework

The framework of FPSO is given in Algorithm 5. FPSO is characterized by firefighting weighted selection and uses the variable length coding method in *Step 7*.

6.2.1. Variable length coding method of aircraft group II

Fig. 6 explains the variable length coding method of aircraft group II. Firstly, the internal fire points only affect the unburned cells inside the fire area. However, the fire edge points may ignite more flammable areas. Therefore, obtain the key rescue cells as a rescue-cell group by using Algorithm 6. Secondly, obtain aircraft group II and *N1* by using Algorithm 7. Lastly, according to the corresponding *AID*, *CID*, and *N1*, we use random number ∈ [0, SRG] to initialize individual coding. One coding corresponds to one individual and also corresponds to the firefighting destination cell of one aircraft. For example, the value of the first gene location in Fig. 6 is 2.1 that is generated by using random number ∈ [0, SRG]. This gene corresponds to the cell with $CID = 11$ and this cell corresponds to the aircraft with $AID = 1$. To simplify the coding complexity of 2-D coordinates (i.e., horizontal

Algorithm 3: Rescue simulator

Input: $N0$, $N1$, $N2$, L^t, XP^t, X^t, YP^t, Y^t, $\boldsymbol{SM}$, T_{rs}, and the current time

Output: the updated parameters (i.e., $N0$, $N1$, $N2$, L^t, XP^t, X^t, YP^t, and Y^t), the updated $\boldsymbol{SM}$ when the current time is an integer multiple of T_{rs}

Step *1*: When the current time is not an integer multiple of T_{rs}, update aircraft groups' members by using Algorithm 2.

Step *2*: When the current time is an integer multiple of T_{rs}, update aircraft groups' members by using Algorithm 2. Then, execute rules *R1*-*R7* as follows. (i) Execute *R1*, *R3*, *R6*, and *R7*, sequentially. (ii) Execute *R2B* and *R5*, when there is aircraft. In this process, each aircraft in aircraft group I uses the firefighting greedy strategy in Algorithm 4. (iii) Execute *R2A* and *R4*.

Algorithm 4: the firefighting greedy strategy of aircraft group I

Step 1: One aircraft arrives at the nearest burning cells in its 3×3 cell neighbor, then it sprays water and puts out the fire of these burning cells.

Step 2: Repeat *Step 1* for its 5×5 cell neighbor, its 7×7 cell neighbor, and so on, until all the water of this aircraft is sprayed.

Algorithm 5: FPSO

Input: NT, $N1$, $N2$, XP^t, X^t, YP^t, Y^t, $\boldsymbol{SM}$, V, T_{ra}, population size PS, the number of generations NG, inertia weight w, acceleration factor 1 c_1, acceleration factor 2 c_2, and particle velocity limit V_{lim}

Output: the aircraft task allocation solution, the updated $N1$, $N2$, XP^t, X^t, YP^t, Y^t, and L^t

Step *1*: In the first run of rescue algorithm, initialize particle position *POS* and particle velocity *VEL* by using a uniformly distributed random generator. Evaluate *POS* and initialize the particle best-known position *pBEST* equal to *POS*. In other runs, *POS* remains unchanged. That is to say, we do not initialize *POS*, except for the first run.

Step 2: Obtain the global best individual *gBEST* by using firefighting weighted selection in Algorithm 8. Then, define the current generation number as *igen* and initialize *igen* equal to 0.

Step 3: Update *VEL* with V_{lim} by using Eq. (9) and *RV* is a random matrix $\in$ [0, 1].

$$VEL \leftarrow w \times VEL + c_1 \times RV \times (pBEST - POS) + c_2 \times RV \times (gBEST - POS) \tag{9}$$

Step 4: Update *POS* by using

$$POS \leftarrow POS + VEL \tag{10}$$

Step 5: Evaluate *POS*. If one position in *POS* is better than its corresponding position in *pBEST*, then update.

Step 6: Update the global best individual *gBEST* by using firefighting weighted selection and then copy *gBEST* into its record *gBESTRECORD*.

Step 7: Compute the current simulation time *CST* by using Eq. (11) and execute Algorithm 7 to update *N1*. If *N1* increases, update *POS*, *VEL*, *pBEST*, and *gBEST*.

$$CST \leftarrow \textit{the beginning time of this period} + igen \times T_{ra}/NG \tag{11}$$

Step 8: If the stop condition of rescue algorithm is not satisfied, go to *Step 3*. Otherwise, use the best individual of *gBESTRECORD* as the aircraft task allocation solution.

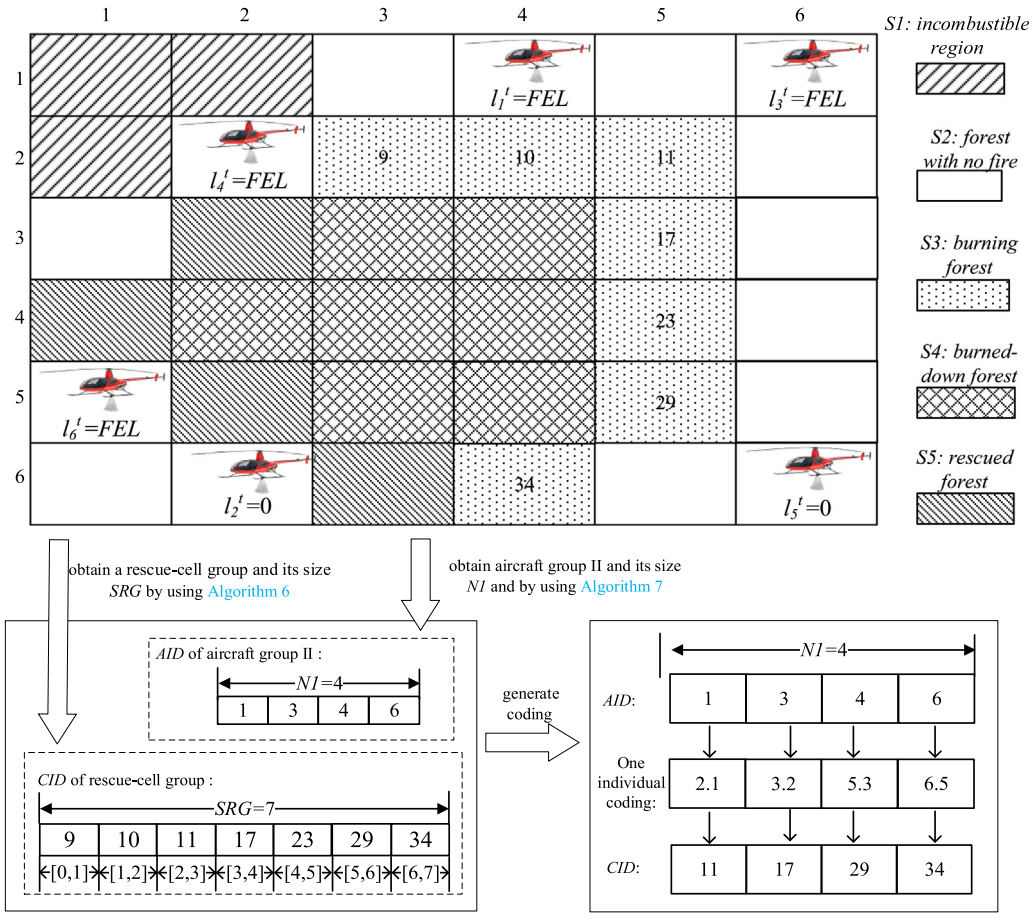


Fig. 6. An illustration of the variable length coding method of aircraft group II in rescue algorithm.

& vertical coordinates), we do not use vertical coordinate and only use the horizontal-center-axis of the x_k^t cell as coding. That is to say, we let the vertical coordinate of x_k^t be always equal to the vertical coordinate of the half height of the x_k^t cell.

6.2.2. *The greedy strategy of aircraft group III to load water*

Aircraft group III uses greedy strategy to load water in Algorithm 7. That is to say, each aircraft in aircraft group III is flying to its nearest firefighting base. We assume that these nearest firefighting bases may be lakes or rivulets. Aircraft are able to load water from these bases at any time, because forest fire has no effect on the use of lakes or rivulets.

6.3. *Firefighting weighted selection and discussions*

Forest firefighting requires the specialist knowledge and background. Therefore, we use firefighting weighted selection to obtain the global best individual *gBEST* in Algorithm 8 and use *gBEST* to guide the population search in Algorithm 5.

By using firefighting weighted selection, we intensify the firefighting forces for the head of forest fire and the continuous-burning-edge regions. From a firefighting perspective, the individual *gBEST* usually includes more genes that aircraft is able to reach its firefighting destination cell or/and includes more genes that aircraft is flying to the head of forest fire & the continuous-burning-edge regions. Explanations are as follows. (i) Eq. (13) uses the greedy strategy to realize controlling forest fire spread immediately. If one individual has more aircraft that are able to reach its firefighting destination cell x_k^t in one period, then it has a higher score. This also means immediately minimizing the newly-increased burnt-forest-cost loss. (ii) From a forest-fire-spread perspective, Eq. (14) emphasizes that aircraft should be allocated to the head of forest fire. The head of forest fire is the most active and fastest-spreading region, because of wind function. When the direction from the initial ignition point to the firefighting destination cell of aircraft is the same with wind direction, this means that the aircraft is flying to the head of forest fire. If there are more aircraft that are flying to the head of forest fire, this means that forest fire spread is more likely to be controlled. (iii) From a burning continuity perspective, Eq. (15) emphasizes that aircraft should be allocated to the important burning regions. For example, if there are more burning cells in one neighbor, this means there are usually more burning cells in the future. Therefore, if more aircraft are allocated to fly to these 3×3 neighbors with more burning cells, then this means that these important burning regions are more likely to be controlled at present or in the future.

On the other hand, Eqs. (13), (14), and (15) use the specialist knowledge of forest fire rescue, which improve the performance of FPSO. However, these knowledge also limits the application scope of FPSO, such as other dynamic programming problems.

7. Experiments and discussions

7.1. *Experimental design*

We wrote all programs in Matlab R2012a that could be downloaded from https://github.com/1654402787/RescueEnsemble and executed these programs on a virtual machine with Windows 7, 2 sub-CPUs of Intel(R) Xeon(R) CPU X5670 @ 2.93 GHz, and 2 GB RAM.

7.1.1. *Setting*

Our benchmarks were based on forest, basin, island, and mountain maps, which could be downloaded from https://github.com/16544027

Algorithm 6: obtain the key rescue cells in forest fire edge as a rescue-cell group

Input: state matrix ***SM***

Output: rescue-cell group and its size *SRG*

1: According to ***SM***, search all burning cells in the forest fire edge.

2: Regard all burning cells in forest fire edge as a rescue-cell group and then obtain its size *SRG*.

Algorithm 7: the greedy strategy of aircraft group III to load water

Input: *CST*, *N1*, *N2*, YP^t, Y^t, and *V*

Output: the updated values of *N1*, *N2*, YP^t, and Y^t

1: According to *CST*, update YP^t first.

2: Check whether aircraft in aircraft group III reach their own nearest firefighting bases or not. If the aircraft

3: arrives at its own firefighting base, then load water and update *N1* & *N2*.

Algorithm 8: firefighting weighted selection

Input: the particle best-known position *pBEST* and their fitness

Output: the global best individual *gBEST*

1: Obtain the 25% best individuals as temporary population *TempPopu* from *pBEST*.

2: Compute the score *SC* of each individual in *TempPopu* by using

3: $$SC = \sum_{k=1}^{N1} IsReach + \sum_{k=1}^{N1} DirectionMatch + \sum_{k=1}^{N1} IsContinuity \quad (12)$$

4:

5: $$IsReach = \begin{cases} 1 & if\ aircraft\ is\ able\ to\ reach\ x_k^t\ in\ this\ T_{ra} \\ 0 & otherwise \end{cases} \quad (13)$$

6: $$DirectionMatch = \cos < \overline{wind\ \ direction}, \overline{direction\ \ from\ initial\ ignition\ \ piont\ to\ aircraft} > \quad (14)$$

7: $$IsContinuity = the\ number\ of\ \ burning\ cells\ in\ the\ 3 \times 3\ neighborhood\ /\ 9 \quad (15)$$

8: According to these *SCs*, we use the highest score individual as *gBEST*.

87/BenchmarksRescueEnsemble. Details were as follows. (i) Vegetation distribution maps were from Google Map. (ii) The angle of wind direction was the angle between north direction and wind. To compare all algorithms objectively, we assumed that wind direction and speed were fixed. (iii) We could not obtain the altitude data for each cell in maps. Therefore, we assumed that $E_1 = E_2$ and obtained $p_s = 1$ (see Section 5.2.1). (iv) Our simulation was offline. To compare all algorithms objectively, we assumed that T_{rs} and T_{ra} were fixed. (v) Aircraft speed *V*. According to real helicopter speeds, $V = 50$ m/s. (vi) *FEC* was decided by *FEL* and the cell's area. The aircraft loads were about from the 1.6 tons of water to the 12 tons of water (see Wangyi, 2019; Sohu, 2019C; People, 2019; Weixin, 2019). $L = 5$ m and cell's area = 25 m^2. According to these data, we assumed $FEC = 8$ that corresponded to 8 cells' area (i.e., 200 m^2). If *FEL* was 12 tons of water, then the load of spray water in 1 m^2 was 60 kg (i.e., 12 tons/200 m^2 = 60 kg/m^2) (see Table 4).

We selected four algorithms as compared rescue algorithms: CMA evolution strategy (CMA-ES) (Varelas et al., 2018; Hansen, 2006), league championship algorithm (LCA) (Kashan, 2014), spider monkey optimization (SMO) (Sharma et al., 2017; Bansal et al., 2014), and a self-adaptive artificial bee colony algorithm (SABC) (Xue et al., 2018). Due to our new benchmarks, there were few compared rescue algorithms for forest fire rescue, as we know. We used these algorithms as compared rescue algorithms because they were the new nature inspired algorithms. Besides, these algorithms used the same methods of the variable length coding of aircraft group II and the greedy strategy of aircraft group III to load water. These were only to adapt to our forest fire rescue problem, which did not change the design details of compared rescue algorithms. On the other hand, we explained the parameter setting of compared rescue algorithms and FPSO as follows. (i) To compare rescue algorithms with FPSO objectively, all algorithms use the same population size and stop condition. (ii) CMA-ES, LCA, SMO, and SABC used the same parameter setting in its own paper (Hansen, 2006; Kashan, 2014; Bansal et al., 2014; Xue et al., 2018), respectively. (iii) FPSO used the common setting of c_1, c_2, and *w*. Besides, V_{lim} was an important parameter to determine the performance of FPSO. Therefore, we provided the sensitivity analysis to explain the V_{lim} setting in Fig. 7. Meanwhile, to express the sensitivity analysis results concisely, we used F1, F3, F5, F7, and F9 as the representatives of our test problems. In Fig. 7, these curves' trend indicates that when V_{lim} is 4, 6, and 8, FPSO is able to obtain better performance. Thus, we set V_{lim} as 8 in this paper (see Table 5).

Table 4
Used parameter setting of rescue ensemble and simulator in this paper.

No.	Name	(latitude, longitude)	Wind		Common parameter	
			Direction	Speed (WV)	Items	Settings
F1	BaDaLing forest	(116.0, 40.4)	0°	10 m/s	p_s	1
F2	White grass forest	(115.5, 39.8)	90°	8 m/s	T_{rs}	20 s
F3	MangShan forest	(116.3, 40.3)	0°	6 m/s	T_{ra}	10 s
F4	LongBeiShan forest	(119.8, 31.3)	90°	6 m/s	T_{res}	10 s
F5	DuHe basin	(110.6, 32.7)	90°	4 m/s	V	50 m/s
F6	XiJiang basin	(108.1, 26.9)	90°	6 m/s	L	5 m
F7	HuaYu island	(128.6, 43.7)	0°	8 m/s	FEC	8
F8	Spetses island	(37.3, 23.1)	180°	10 m/s	NT	10
F9	HuaShan mountain	(110.1, 34.5)	0°	6 m/s	R	300
F10	TaiShan mountain	(117.1, 36.3)	135°	8 m/s	C	300

Table 5
Used parameter setting of compared rescue algorithms and proposed FPSO in this paper.

Item	Parameter
CMA-ES	Parameter setting of CMA-ES in this paper was the same with Hansen (2006). We did not provide these detailed values of CMA-ES, because of the size of parameter settings. However, these values and source code of CMA-ES could be found in Hansen (2006).
LCA	Population size $PS = 80$, the retreat contribution coefficient $\psi_1 = 0.5$, the approach contribution coefficient $\psi_2 = 1$, the probability of success in the truncated geometric distribution $p_c = 0.5$
SMO	Population size $PS = 80$, $MG = 5$, $GlobalLeaderLimit = 50$, $LocalLeaderLimit = 1500$, pr increases over iterations from 0.1 to 0.4, linearly.
SABC	Population size $PS = 80$, the value of limit $limit = 0.6\times(PS/2)\times N1$, the fixed number of previous generations $LP = 10$
FPSO	Population size $PS = 80, c_1 = 2, c_2 = 2, V_{lim} = 8$, with generations increasing, w decreases from 1 to 0.2
Stop condition	Stop condition of each rescue algorithm is that the number of evaluation times was 4000.

7.1.2. Objectives

In Section 7.2, dynamic search experiments used the offline error rate μ as the evaluation criterion. The smaller μ means that rescue algorithm has better performance to obtain the best solution in the dynamic fitness landscape, continuously. The offline error rate μ (Yang and Li, 2010) was defined by

$$\mu = \frac{1}{K}\sum_{k=1}^{K}|h_k - f_k| \tag{16}$$

where K was the number of dynamic fitness landscape changes, f_k was the optimal solution obtained by the algorithm from the *1*st fitness landscape to the *k*th fitness landscape, and h_k was the optimal value of the algorithm in the *k*th fitness landscape.

In Section 7.3, forest-fire-rescue experiments used evaluation criteria as follows. (i) *FT* was equal to the number of performing rescue algorithm $\times T_{ra}$. The smaller *FT*, the better performance. (ii) *RER* was equal to the number of the total cells that have been put out in the edge after fire (*PutOutNumberInEdge*) / the number of the total burnt cells after fire (*BurntCellNumber*) in our map. Note that, the greater *RER* corresponds to the greater *PutOutNumberInEdge* and the smaller *BurntCellNumber*. Therefore, the greater *RER*, the better performance. (iii) *BVR* was equal to the cost of the total burned cells after fire/the total cost of cells in our map. The smaller *BVR*, the better performance.

7.2. Dynamic search results

To demonstrate the dynamic search performance of FPSO, Fig. 8 provided the mean value of μ for 30 runs. Except for F5, F6, F8, and F10, FPSO had the best results. This indicates that FPSO is able to track the best solution, even in the dynamic forest-fire-spread environments. There were 300 tests of F1–F10 and these 300-test results explained that the dynamic search capability of FPSO was suitable for minimizing the longest flight time and the forest cost loss of the newly-increased burnt cells, such as adapting to the time-varying shape of forest fire, the various kinds of maps (like various vegetation distributions). Generally, this indicates that the dynamic convergence of FPSO is stable in most cases.

To further demonstrate the dynamic search capability of FPSO, we used Wilcoxon test at 0.05 level of significance for the 30-run μ data sets of two algorithms, as shown in Table 6. Wilcoxon test was widely used to compare/analyze the difference between two evolutionary algorithms. From a dynamic convergence perspective, the symbols of (+), (≈), and (−) mean that FPSO is better than, approximately equal to, and worse than the compared rescue algorithm, respectively. *RSum*(+) and *RSum*(−) return the numbers of plus sign (+) and minus sign (−) in a row, respectively. *RSum*(+) was always greater than *RSum*(−). This indicates that FPSO is better than CMA-ES, LCA, SMO, and SABC. In addition, *CSum*(+) and *CSum*(−) return the numbers of plus sign (+) and minus sign (−) in a column, respectively. Except for F6 and F8, *CSum*(+) was greater than *CSum*(−). This means that the search capability of FPSO is satisfactory when applying to dynamic forest-fire-spread environments.

Table 6
Wilcoxon results of the offline error rate μ between one compared rescue algorithm and FPSO.

Algorithm	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	*RSum*(+)	*RSum*(−)
CMA-ES	+	+	≈	+	≈	≈	+	≈	≈	+	5	0
LCA	+	≈	≈	≈	≈	≈	+	−	+	≈	3	1
SMO	+	≈	+	≈	+	≈	≈	≈	+	≈	4	0
SABC	+	≈	≈	+	≈	≈	+	≈	+	≈	4	0
CSum(+)	4	1	1	2	1	0	3	0	3	1	*NULL*	*NULL*
CSum(−)	0	0	0	0	0	0	0	1	0	0	*NULL*	*NULL*

7.3. Forest-fire-rescue results

In real forest-fire-rescue environments, command staff pay more attention to the specific effects of fire rescue, such as rescue time, fire edge suppression, and burned cost. In destructive forest fire disasters, these mean decreasing casualties and losses. To compare with the key rescue effects, we provided *FT*, *RER* and *BVR* results as follows.

7.3.1. Forest-fire-rescue time

FT is one of the most important rescue indexes from a real rescue perspective, and the reason is as follows. *FT* not only is directly proportional to the capability of controlling forest fire spread but also

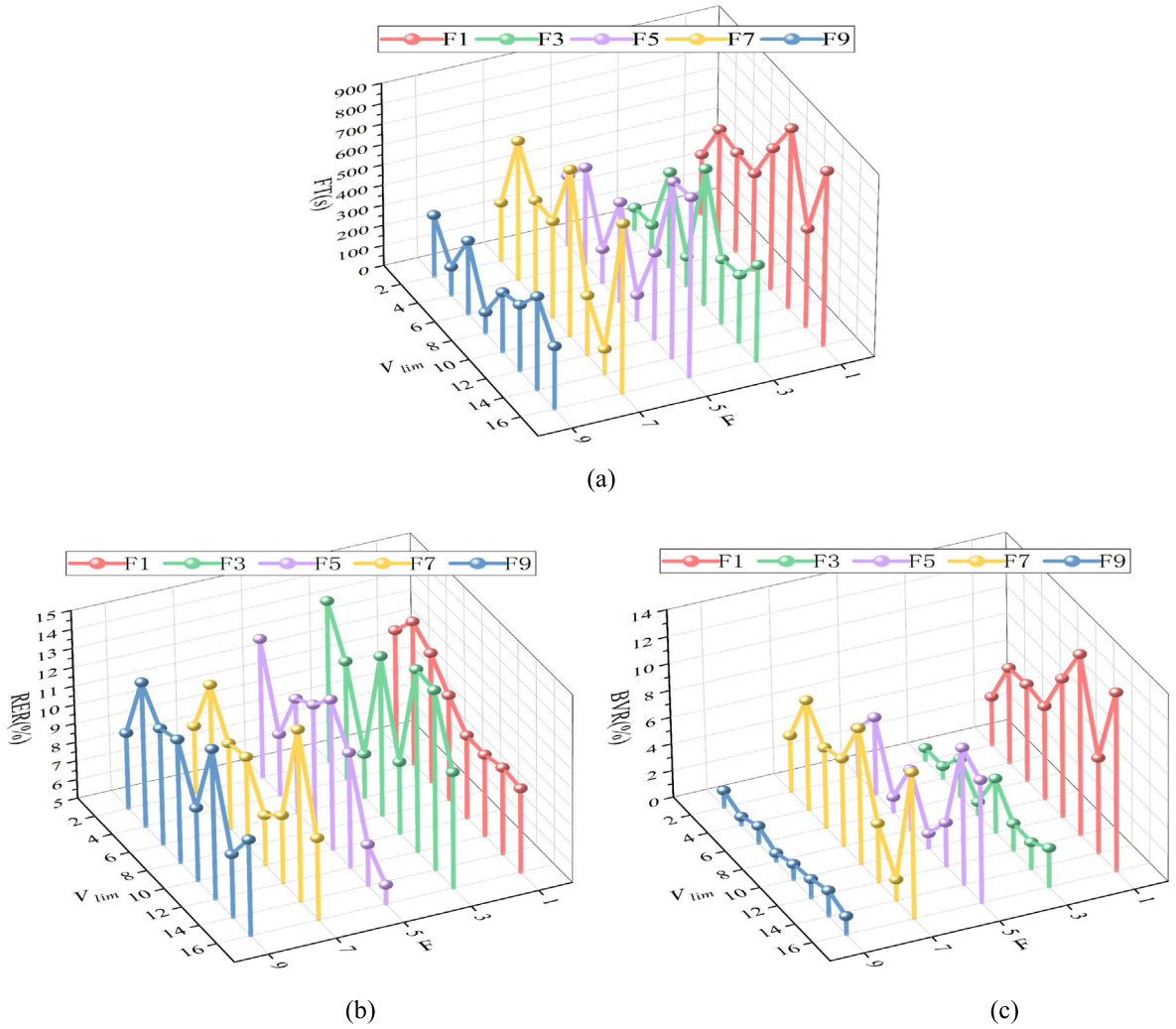


Fig. 7. Sensitivity analysis results to explain the V_{lim} setting in FPSO by using the mean of the 10-run *FT*, *RER*, and *BVR* data.

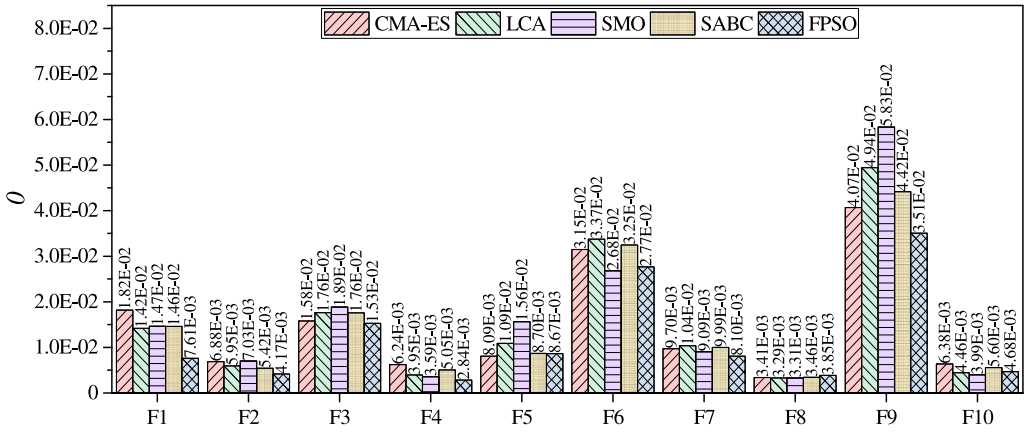


Fig. 8. The mean value of the offline error rate μ for 30 runs to test the dynamic search performance of FPSO.

corresponds to reducing the risk that large-scale forest fire happens. Fig. 9 provided the mean value of *FTs* for 30 runs. Except for F5 and F10, FPSO had the best results.

To further demonstrate the FPSO performance in respect of putting out forest fire as soon as possible, we used Wilcoxon test at 0.05 level of significance, and the results were given in Table 7. Wilcoxon results of FPSO were the best. Comparing with Table 6, the performance of FPSO in Table 7 for F6 and F8 was contrary to Table 6. The reason is that the results in Table 6 are to simultaneously minimize the longest flight time and the forest cost loss of the newly-increased burnt cells. This was different from the objective of putting out forest fire as soon as possible in Table 7. These *FT* results of FPSO were due to firefighting weighted selection, which considered the aircraft reachability, wind direction, and burning continuity. Meanwhile, these *FT* results indicate

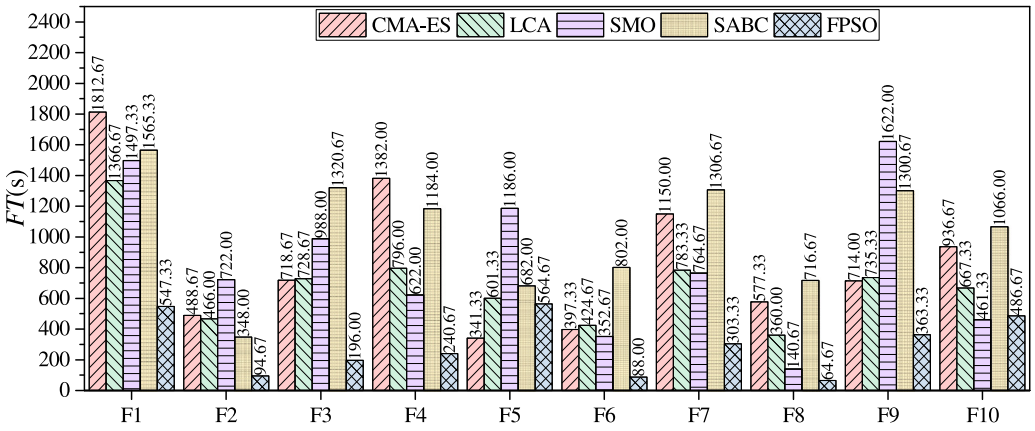


Fig. 9. The mean value of *FTs* for 30 runs to illustrate the performance of FPSO in respect of putting out forest fire as soon as possible.

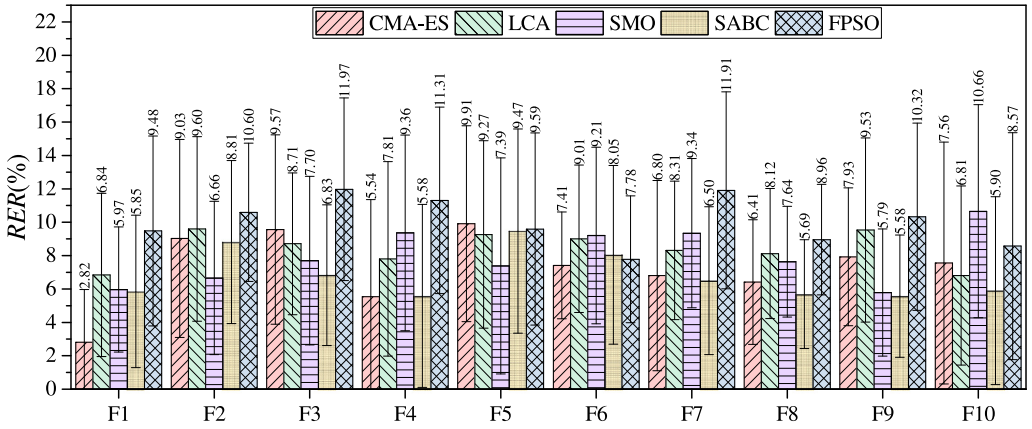


Fig. 10. *RER* results about the mean value and standard deviation for 30 runs to illustrate the performance of FPSO in respect of controlling the spread speed of forest fire edge as much as possible.

Table 7
Wilcoxon results of *FTs* between one compared rescue algorithm and FPSO.

Algorithm	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10
CMA-ES	+	+	+	+	−	≈	+	+	≈	+
LCA	+	≈	+	≈	≈	+	≈	+	≈	≈
SMO	+	+	+	≈	+	+	≈	≈	+	≈
SABC	+	≈	+	+	≈	+	+	+	+	+
CSum(+)	4	2	4	2	1	3	2	3	2	2
CSum(−)	0	0	0	0	1	0	0	0	0	0

that considering the aircraft reachability, wind direction, and burning continuity is suitable for improving the forest-fire-rescue effects.

7.3.2. Rescued edge ratio

Although the area of forest fire edge is far less than the internal region of forest fire, reducing the spread speed of forest fire edge is the most effective firefighting method in real forest fires. Moreover, spraying water in the internal region of forest fire is useless to put out forest fires.

Fig. 10 provided the mean value and standard deviation of *RERs* for 30 runs. Expect for F5, F6, and F10, *RER* results of FPSO were the best from a mean-value perspective. In F5 and F6, FPSO was the second. In FPSO, we improved and strengthened the strategy of putting out the fire in the head of forest fire by using Eq. (14). The head of forest fire was the fastest-spreading region in forest fire edge, which was also the most harmful part to endanger personal safety. In most cases, the task that ground firemen implement firefighting in the head of forest fire was forbidden (Xiao et al., 2006), especially when wind speed was high. However, aircraft was flying in the sky, which was not constrained by the previous forest-fire-rescue standard for ground firemen (Xiao et al., 2006). These *RER* results indicate that the strategy of using aircraft to extinguish the head of forest fire continuously is effective.

7.3.3. Burned value ratio

The cost value of burned forest not only reflects the effects of controlling the spread speed of forest fire edge but also is related with the density of forest vegetation of burned cells. Fig. 11 gave *BVR* results to illustrate the cost value of the burned forest. We computed the mean value of *BVRs* for 30 runs. Except for F5 and F10, FPSO had the best results. In F5 and F10, FPSO was the second. Besides, the *BVR* distributions of FPSO were closer to smaller values (i.e., the bottom of the boxes) in most cases. This indicates that the FPSO's capability of controlling the burned-forest cost is satisfactory.

8. Conclusions

Experiments demonstrate that rescue ensemble is a feasible framework to realize the dynamic rescue process simulation between forest fire spread and forest fire rescue. Meanwhile, experimental results demonstrate that the dynamic search performance of FPSO is suitable for realizing aircraft task allocation and the adaptability of FPSO is mainly from using the firefighting specialist knowledge, such as considering the aircraft reachability, controlling the spread speed of the head of forest fire, enhancing the rescue effort on the burning-continuity-edge region. Besides, these results indicate that various rescue algorithms lead to different rescue effects. However, when we

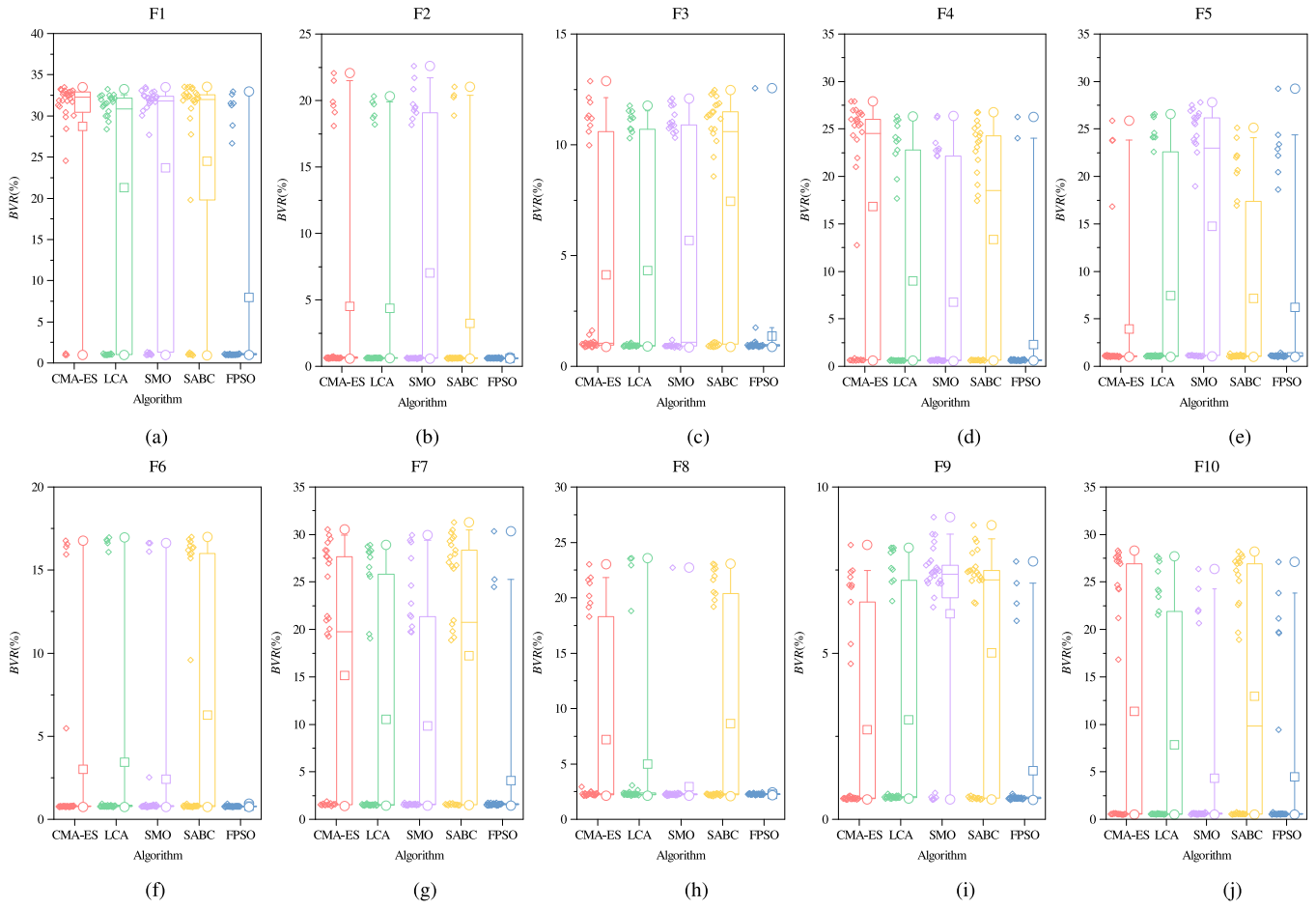


Fig. 11. *BVR* results for 30 runs to illustrate the performance of FPSO in respect of controlling the cost of burned forest.

have the same limited rescue aircraft in the early-stage forest fire, the unsatisfactory effect of one rescue algorithm means that much more time and effort are required to control the continuously growing forest fire in the following stage.

Our research is limited to the early-stage forest fire rescue by using firefighting aircraft in this paper. With the new unmanned platforms increasing, forest-fire-rescue problems with ground and air firefighting equipments are interesting researches in the forest firefighting field.

CRediT authorship contribution statement

HongGuang Zhang: Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **ZiHan Liang:** Software, Data curation, Visualization, Investigation, Validation. **HuaJian Liu:** Software, Data curation, Visualization, Investigation, Validation. **Rui Wang:** Software, Data curation, Visualization, Investigation, Validation. **YuanAn Liu:** Software, Data curation, Visualization, Investigation, Validation.

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